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$f(x, y, z) = e^{xyz}$ in $a(4, 0, -3)$ in de richting $h(0, 1, -1)$

$$f(4, 0, -3) = 1$$

$$\begin{aligned} Df(a, h) &= \lim_{\lambda \rightarrow 0} \frac{f((4, 0, -3) + \lambda(0, 1, -1)) - 1}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} \frac{e^{-4\lambda^2 - 12\lambda}}{\lambda} \\ &= \lim_{\lambda \rightarrow 0} (-8\lambda - 12) e^{-4\lambda^2 - 12\lambda} \\ &= -12 \end{aligned}$$

References